A RESTRICTED FLAT, LOCALLY FINITELY PRESENTED PRE-RELATION HAS OPEN QUOTIENT MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains two xymatrix displays; they are part of the counted proof span and are preserved verbatim, with the chapter's own Xy-pic setup retained in the preamble.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Restriction of an equivalence relation along a morphism is an equivalence relation. Groupoids, Lemma groupoids-lemma-restrict-relation, used at proof lines 1544-1546. Gives the equivalence-relation hypothesis for R'.
- Flat and locally finitely presented morphisms are universally open. Morphisms, Lemma morphisms-lemma-fppf-open, used at proof lines 1546-1549. Makes W^1 and W open in U.
- The invariant open attached to an equivalence relation. Groupoids, Lemma groupoids-lemma-constructing-invariant-opens, used at proof lines 1550-1554. Supplies t(s^{-1}(W)) = W.
- Injectivity of U'/R' -> U/R for the restricted relation. Groupoids, Lemma groupoids-lemma-quotient-pre-equivalence-relation-restrict, used at proof lines 1557-1560. Turns the claim into representing the fibre product by an open subscheme.
- Openness descends through an fpqc covering. Descent, Lemma descent-lemma-open-fpqc-covering, used at proof lines 1594-1597. Produces the open W_T subset T with phi_j^{-1}(W_T) = W_j.

Source reference. Original lemma 03N0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4768–4768. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03LO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4720–4720. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 045Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5454–5454. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02V8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3759–3759. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01UA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2403–2403. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let U be a scheme over S. Let $j = (s,t) : R \to U \times_S U$ be a pre-relation. Let $g : U' \to U$ be a morphism. Assume*

(1) *j is an equivalence relation,*
(2) *$s, t : R \to U$ are surjective, flat and locally of finite presentation,*
(3) *g is flat and locally of finite presentation.*

Let $R' = R|_{U'}$ be the restriction of R to U'. Then $U'/R' \to U/R$ is representable, and is an open immersion.

Proof. By Groupoids, Lemma 02V8 the morphism $j' = (s', t') : R' \to U' \times_S U'$ defines an equivalence relation. Since g is flat and locally of finite presentation we see that g is universally open as well (Morphisms, Lemma 01UA). For the same reason s, t are universally open as well. Let $W^1 = g(U') \subset U$, and let $W = t(s^{-1}(W^1))$. Then W^1 and W are open in U. Moreover, as j is an equivalence relation we have $t(s^{-1}(W)) = W$ (see Groupoids, Lemma 03LO for example).

By Groupoids, Lemma 045Z the map of sheaves $F' = U'/R' \to F = U/R$ is injective. Let $a : T \to F$ be a morphism from a scheme into U/R. We have to show that $T \times_F F'$ is representable by an open subscheme of T.

The morphism a is given by the following data: an fppf covering $\{\varphi_j : T_j \to T\}_{j \in J}$ of T and morphisms $a_j : T_j \to U$ such that the maps

$$a_j \times a_{j'} : T_j \times_T T_{j'} \longrightarrow U \times_S U$$

factor through $j : R \to U \times_S U$ via some (unique) maps $r_{jj'} : T_j \times_T T_{j'} \to R$. The system (a_j) corresponds to a in the sense that the diagrams

$$\begin{array}{ccc} T_j & \xrightarrow{a_j} & U \\ \downarrow & & \downarrow \\ T & \xrightarrow{a} & F \end{array}$$

commute.

Consider the open subsets $W_j = a_j^{-1}(W) \subset T_j$. Since $t(s^{-1}(W)) = W$ we see that

$$W_j \times_T T_{j'} = r_{jj'}^{-1}(t^{-1}(W)) = r_{jj'}^{-1}(s^{-1}(W)) = T_j \times_T W_{j'}.$$

By Descent, Lemma 03N0 this means there exists an open $W_T \subset T$ such that $\varphi_j^{-1}(W_T) = W_j$ for all $j \in J$. We claim that $W_T \to T$ represents $T \times_F F' \to T$.

First, let us show that $W_T \to T \to F$ is an element of $F'(W_T)$. Since $\{W_j \to W_T\}_{j \in J}$ is an fppf covering of W_T, it is enough to show that each $W_j \to U \to F$

is an element of $F'(W_j)$ (as F' is a sheaf for the fppf topology). Consider the commutative diagram

$$\begin{array}{ccccc}
W'_j & & \longrightarrow & & U' \\
\downarrow & \searrow & & & \downarrow{\scriptstyle g} \\
 & & s^{-1}(W^1) & \xrightarrow[s]{} & W^1 \\
 & & \downarrow{\scriptstyle t} & & \downarrow \\
W_j & \xrightarrow{a_j|_{W_j}} & W & \longrightarrow & F
\end{array}$$

where $W'_j = W_j \times_W s^{-1}(W^1) \times_{W^1} U'$. Since t and g are surjective, flat and locally of finite presentation, so is $W'_j \to W_j$. Hence the restriction of the element $W_j \to U \to F$ to W'_j is an element of F' as desired.

Suppose that $f : T' \to T$ is a morphism of schemes such that $a|_{T'} \in F'(T')$. We have to show that f factors through the open W_T. Since $\{T' \times_T T_j \to T'\}$ is an fppf covering of T' it is enough to show each $T' \times_T T_j \to T$ factors through W_T. Hence we may assume f factors as $\varphi_j \circ f_j : T' \to T_j \to T$ for some j. In this case the condition $a|_{T'} \in F'(T')$ means that there exists some fppf covering $\{\psi_i : T'_i \to T'\}_{i \in I}$ and some morphisms $b_i : T'_i \to U'$ such that

$$\begin{array}{ccccc}
T'_i & \xrightarrow[b_i]{} & U' & \xrightarrow[g]{} & U \\
{\scriptstyle f_j \circ \psi_i}\downarrow & & & & \downarrow \\
T_j & \xrightarrow{a_j} & U & \longrightarrow & F
\end{array}$$

is commutative. This commutativity means that there exists a morphism $r'_i : T'_i \to R$ such that $t \circ r'_i = a_j \circ f_j \circ \psi_i$, and $s \circ r'_i = g \circ b_i$. This implies that $\mathrm{Im}(f_j \circ \psi_i) \subset W_j$ and we win. □